

Rank-metric intersecting codes

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Abstract

Intersecting codes are codes in which any two nonzero codewords share a common nonzero coordinate, and they have been extensively studied due to their applications in communication theory, cryptography, and combinatorics. Recently, geometric interpretations have provided new insights into intersecting codes, both in the Hamming metric and, more recently, in the rank metric for linear codes.

In this paper, we introduce and study matrix rank-metric intersecting codes. By defining the support of a matrix as its row space, we extend the notion of intersecting codes to the matrix rank-metric setting. Using the slice space associated with the generator tensor of a rank-metric code, we derive a geometric characterization that is equivalent to the intersecting property. This characterization generalizes the known results for linear rank-metric intersecting codes and highlights both analogies and differences with the linear case. Our approach provides a new geometric framework for the study of intersecting codes in the rank metric and opens the way to further investigations of their structural and extremal properties.

Key Words : Rank-metric codes, intersecting codes, tensors.

Introduction

Intersecting codes, first introduced in [KS83, Mik84], are codes for which every two nonzero codewords have a nonzero coordinate in common. They

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have numerous applications, for instance for communication over an AND channel, for digital fingerprinting [SSW01, Bla03], for oblivious transfer [BCS96, BCS02], and they are related to several combinatorial structures such as $(2, 1)$ -separating systems [Ran13]. In the binary case, they coincide with minimal codes, which have also been intensively studied [AB98, CMP13, ABN22], and have been used by Massey to develop a secret sharing scheme [Mas93]. The theory of intersecting codes was further developed in various works such as [Ret89, CZ94, CELS03, Ran13]. More recently, connections with additive combinatorics and factorization theory have also been explored [PS10, BSS25]. In [BSS25], the authors provide a geometric interpretation of intersecting codes in the Hamming metric: intersecting codes correspond to sets of points in the projective space that are not contained in the union of two hyperplanes. This allows various geometric investigations of intersecting codes.

In the last two decades, rank-metric codes have become a central subject in coding theory, motivated by applications in areas such as network coding and post-quantum cryptography, as well as by their strong links with finite geometry. The geometric interpretation of linear rank-metric codes through q -systems [Ran20] has proved to be a powerful tool for studying extremal classes of codes, including MRD and minimal linear rank-metric codes, and related structures [She19, ZZ21, ABNR22]. Linear intersecting codes in the rank metric have been introduced in [BBMS25] with the authors providing a geometric interpretation of intersecting codes in terms of q -systems, together with bounds on the parameters, constructions, and relations with other combinatorial structures. Finally, in a paper under preparation [ABN26], the first author **et al.** are investigating the geometry of matrix rank-metric codes.

Building on these previous works, our aim is to propose a definition of matrix rank-metric intersecting codes. In analogy with [BBMS25], we will define the support of a matrix to be its rowspan (it is also possible to consider the columnspan but this yields a similar theory by transposition). Examining the slice space derived from the generator tensor of our rank-metric code, we derive a property for this slice space that is equivalent to the code being intersecting. This geometric property specializes to the one developed in [BBMS25] in the linear case. We then derive geometric properties of the slice spaces corresponding to intersecting codes, pointing out both the differences and the similarities to [BBMS25].

Our work is organized as follows: in Section 1, we provide a rapid introduction of rank-metric codes and tensors, and we introduce intersecting codes in this context. In Section 2, using the new geometric interpretation of matrix rank-metric codes developed in [ABN26], we propose a geometric interpretation of rank-metric intersecting codes.

1 Preliminaries and first properties

We consider tensor products of the form $\mathbb{F}_q^{N_1} \otimes \mathbb{F}_q^{N_2} \otimes \mathbb{F}_q^{N_3}$, whose elements are called 3-tensors. The elements of this space can be represented as 3-dimensional arrays. We can identify $\mathbb{F}_q^{N_1} \otimes \mathbb{F}_q^{N_2} \otimes \mathbb{F}_q^{N_3}$ with $\mathbb{F}_q^{N_1 \times N_2 \times N_3}$. We introduce the following maps:

$$\begin{aligned} m_1 : \mathbb{F}_q^{N_1 \times N_2 \times N_3} \times \mathbb{F}_q^{N_1 \times s} &\rightarrow \mathbb{F}_q^{s \times N_2 \times N_3} \\ (T, A_1) &\mapsto m_1(T, A_1) := \sum_i (u_i A_1) \otimes v_i \otimes w_i, \\ m_2 : \mathbb{F}_q^{N_1 \times N_2 \times N_3} \times \mathbb{F}_q^{N_2 \times s} &\rightarrow \mathbb{F}_q^{N_1 \times s \times N_3} \\ (T, A_2) &\mapsto m_2(T, A_2) := \sum_i u_i \otimes (v_i A_2) \otimes w_i, \\ m_3 : \mathbb{F}_q^{N_1 \times N_2 \times N_3} \times \mathbb{F}_q^{N_3 \times s} &\rightarrow \mathbb{F}_q^{N_1 \times N_2 \times s} \\ (T, A_3) &\mapsto m_3(T, A_3) := \sum_i u_i \otimes v_i \otimes (w_i A_3), \end{aligned}$$

for any $T = \sum_i u_i \otimes v_i \otimes w_i \in \mathbb{F}_q^{N_1 \times N_2 \times N_3}$.

Definition 1.1. Let $T \in \mathbb{F}_q^{N_1 \times N_2 \times N_3}$. For each $i \in \{1, 2, 3\}$, we define the *i-th slice space* of T to be the $\mathbb{F}_q$ -span of $\{m_i(T, e_j) : 1 \leq j \leq N_i\}$, that is,

$$\text{ss}_i(T) := \langle m_i(T, e_1), \dots, m_i(T, e_{N_i}) \rangle_{\mathbb{F}_q}.$$

Definition 1.2. A $[m \times n, k, d]_q$ **rank-metric code** $\mathcal{C}$ is a subspace of $\mathbb{F}_q^{m \times n}$ of **dimension** k and **minimum distance**

$$d := \min\{\text{rk}(A - B) : A, B \in \mathcal{C}, A \neq B\} = \min\{\text{rk}(A) : A \in \mathcal{C}, A \neq 0\}.$$

A **generator tensor** for a code $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ is an element $T \in \mathbb{F}_q^{k \times m \times n}$ such that $\mathcal{C} = \text{ss}_1(T)$. The **dual** of a rank-metric code $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ is the rank-metric code

$$\mathcal{C}^\perp := \{B : \text{Tr}(BA^\top) = 0, \forall A \in \mathcal{C}\}.$$

Remark 1.1. An $\mathbb{F}_q$ -**linear isometry** (with respect to the rank) of $\mathbb{F}_q^{m \times n}$ is a map $\varphi_{M,N} : \mathbb{F}_q^{m \times n} \rightarrow \mathbb{F}_q^{m \times n}$ sending A to MAN , where $M \in \text{GL}_m(\mathbb{F}_q)$ and $N \in \text{GL}_n(\mathbb{F}_q)$. If $n = m$, transposition is also an isometry, but we avoid using that term in order to prevent confusion later.

Definition 1.3. Two $[m \times n, k, d]_q$ rank-metric code $\mathcal{C}, \mathcal{C}'$ are **equivalent** if there is an $\mathbb{F}_q$ -linear isometry sending $\mathcal{C}$ to $\mathcal{C}'$.

Definition 1.4. The **row-support** (respectively the **column-support**) of a matrix $A \in \mathbb{F}_q^{m \times n}$ is $\sigma_r(A) := \text{rowspan}(A)$ (respectively $\sigma_c(A) := \text{colspan}(A)$).

Definition 1.5. A code $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ is **row-intersecting** (respectively **column-intersecting**) if every pair of nonzero **codewords** $A, B \in \mathcal{C}$ has non-trivially intersecting row-supports (respectively column-supports), that is for all $A, B \in \mathcal{C} \setminus \{0\}$, $\sigma_r(A) \cap \sigma_r(B) \neq \{0\}$ (respectively $\forall A, B \in \mathcal{C} \setminus \{0\}$, $\sigma_c(A) \cap \sigma_c(B) \neq \{0\}$).

Remark 1.2. Being intersecting is invariant under equivalence. Moreover, the transposition sends row-intersecting codes to column-intersecting ones.

Example 1.1. The $[4 \times 3, 8, 2]_2$ code generated by the following matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

is row-intersecting but not column-intersecting. The $[4 \times 3, 4, 3]_2$ code generated by the first four matrices is both row-intersecting and column-intersecting. Both codes are Gabidulin codes.

Proposition 1.1. Let $\mathcal{C}$ be an $[m \times n, k, d]_q$ code. If $2d > n$, then $\mathcal{C}$ is row-intersecting. If $2d > m$, then $\mathcal{C}$ is column-intersecting.

Remark 1.3. Note that Proposition 1.1 allows constructions of rank-metric intersecting codes: in fact, it is enough to consider known constructions of MRD codes (Gabidulin codes or their generalization, see for example [LTZ18]) with parameters satisfying the hypothesis of the proposition. Let us also remark that this gives examples of rank-metric intersecting codes which are not necessarily linear.

Definition 1.6. A rank-metric code $\mathcal{C} \in \mathbb{F}_q^{m \times n}$ is said **Hamming row-intersecting** (respectively **Hamming column-intersecting**) if for any $A, B \in \mathcal{C} \setminus \{0\}$, there exists a coordinate i for which the i -th rows (respectively the i -th columns) of A and B are both nonzero.

Theorem 1.1. A rank-metric code $\mathcal{C}$ is row-intersecting (respectively column-intersecting) if and only if, for all $N \in \text{GL}_n(\mathbb{F}_q)$ (respectively $M \in \text{GL}_m(\mathbb{F}_q)$), $\varphi_{I_m, N}(\mathcal{C})$ (respectively $\varphi_{M, I_n}(\mathcal{C})$) is Hamming column-intersecting (respectively Hamming row-intersecting).

Proof. Assume that for some $N \in \text{GL}_n(\mathbb{F}_q)$, the code $\varphi_{I_m, N}(\mathcal{C})$ is not Hamming column-intersecting. Then there exist 2 codewords A, B such that AN and BN do not share a nonzero column. Hence $\sigma_r(AN) \cap \sigma_r(BN) = \{0\}$, which implies $\sigma_r(A) \cap \sigma_r(B) = \{0\}$. Therefore the code $\mathcal{C}$ is not row-intersecting.

Conversely, assume that there exist two codewords $A, B \in \mathcal{C}$ such that $\sigma_r(A) \cap \sigma_r(B) = \{0\}$. Then there exists $N \in \text{GL}_n(\mathbb{F}_q)$ such that AN and BN has no coordinate i for which both AN and BN have a zero column. Hence $\varphi_{I_m, N}(\mathcal{C})$ is not Hamming column-intersecting. $\square$

2 The third slice space of a row-intersecting code

We will now concentrate on row-intersecting codes. We do this for the sake of brevity, and an entirely analogous argument applies to column-intersecting codes as well.

For a code $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ of dimension k , generated by T , let us consider

$$\mathcal{X} = \text{ss}_3(T) \subseteq \mathbb{F}_q^{k \times m}.$$

From now on, we will assume, without loss of generality, that $\dim \mathcal{X} = n$ (which means that the code $\mathcal{C}$ is column-nondegenerate): if this is not the case, $\mathcal{C}$ can be isometrically embedded in a space of matrices $\mathbb{F}_q^{m \times n'}$, with $n' \leq n$, such that $\dim \mathcal{X} = n'$.

Definition 2.1. Let $l \in \{1, \dots, k-1\}$, $r \in \{1, \dots, m-1\}$, $L \in \mathbb{F}_q^{l \times k}$ of full rank, $R \in \mathbb{F}_q^{m \times r}$ of full rank. The **left puncturing** of $\mathcal{X}$ with respect to L is

$${}_L\mathcal{X} = \{LM : M \in \mathcal{X}\} \subseteq \mathbb{F}_q^{l \times m}.$$

The **right puncturing** of $\mathcal{X}$ with respect to R is

$$\mathcal{X}_R = \{MR : M \in \mathcal{X}\} \subseteq \mathbb{F}_q^{k \times r}.$$

Moreover, for a subspace $S \subseteq \mathbb{F}_q^k$, we define

$$\mathcal{X}_c(S) := \{M \in \mathcal{X} : \text{colspan}(M) \subseteq S\}$$

and for a subspace $S \subseteq \mathbb{F}_q^m$, we define

$$\mathcal{X}_r(S) := \{M \in \mathcal{X} : \text{rowspan}(M) \subseteq S\}$$

The **left-shortening** of $\mathcal{X}$ with respect to L is ${}_L\mathcal{X}_c(\text{rowspan}(L)) \subseteq \mathbb{F}_q^{l \times m}$.

The **right-shortening** of $\mathcal{X}$ with respect to R is $\mathcal{X}_r(\text{colspan}(L))_R \subseteq \mathbb{F}_q^{k \times r}$.

Definition 2.2. We say that a (slice) space is **column-hyperplane-decomposable** if there exist two hyperplanes H_1, H_2 of $\mathbb{F}_q^k$ such that

$$\mathcal{X} = \mathcal{X}_c(H_1) + \mathcal{X}_c(H_2).$$

We say that a slice space is **column-hyperplane-indecomposable** if it is not hyperplane-decomposable.

Theorem 2.1. Let $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ be a rank-metric code of dimension k with generator tensor T and let $\mathcal{X} = \text{ss}_3(T) \subseteq \mathbb{F}_q^{k \times m}$ be the third slice space of T . Then

$$\mathcal{C} \text{ is row-intersecting} \iff \mathcal{X} \text{ is column-hyperplane-indecomposable.}$$

Proof. As shown in [ABN26], for all $u \in \mathbb{F}_q^k$,

$$\sigma_r(m_1(T, u))^\perp = \psi^{-1}(\mathcal{X}_c(\langle u \rangle_{\mathbb{F}_q}^\perp)) \quad (1)$$

where

$$\psi : \mathbb{F}_q^n \rightarrow \mathcal{X}, \quad v \mapsto m_3(T, v).$$

Here the duality is with respect to the standard inner product in the corresponding spaces. The slice space $\mathcal{X}$ is column hyperplane-decomposable if and only if there exists $u, u' \in \mathbb{F}_q^k \setminus \{0\}$ such that

$$\mathcal{X} = \mathcal{X}_c(\langle u \rangle_{\mathbb{F}_q}^\perp) + \mathcal{X}_c(\langle u' \rangle_{\mathbb{F}_q}^\perp).$$

This is equivalent to

$$\mathbb{F}_q^n = \psi^{-1}(\mathcal{X}_c(\langle u \rangle_{\mathbb{F}_q}^\perp)) + \psi^{-1}(\mathcal{X}_c(\langle u' \rangle_{\mathbb{F}_q}^\perp)) = \sigma_r(m_1(T, u))^\perp + \sigma_r(m_1(T, u'))^\perp,$$

which happens if and only if

$$\{0\} = (\sigma_r(m_1(T, u))^\perp + \sigma_r(m_1(T, u'))^\perp)^\perp = \sigma_r(m_1(T, u)) \cap \sigma_r(m_1(T, u')).$$

The last condition means that $\mathcal{C}$ is not row-intersecting. $\square$

Proposition 2.1. *If, for every hyperplane $H \subseteq \mathbb{F}_q^k$,*

$$\dim \mathcal{X}_c(H) < \frac{1}{2} \cdot \dim \mathcal{X},$$

then $\mathcal{X}$ is column-hyperplane-indecomposable.

Theorem 2.2. *If $\dim \mathcal{X} \leq 2\lfloor (k-1)/m \rfloor$, then $\mathcal{X}$ is column-hyperplane-decomposable.*

Proof. Let $X_1, \dots, X_n$ be an $\mathbb{F}_q$ -linear basis of the slice space $\mathcal{X}$. For each $1 \leq i \leq n$ write $V_i := \text{colspan}(X_i)^\perp$, which is a vector subspace of $\mathbb{F}_q^k$ of codimension at most m . Noting $j = \lfloor \frac{k-1}{m} \rfloor$, we have $W_1 = V_1 \cap \dots \cap V_j \neq \{0\}$. Indeed, the codimension of W_1 is at most $mj \leq k-1$ by the Grassmann's formula. Therefore there exists a nonzero vector $w_1 \in W_1 \setminus \{0\}$. Similarly, we note $W_2 = V_{j+1} \cap \dots \cap V_n$. Since $n = \dim \mathcal{X} \leq 2j$, we also have $W_2 \neq \{0\}$, so that we may also choose a nonzero vector $w_2 \in W_2 \setminus \{0\}$.

Defining the hyperplanes $H_1 = \langle w_1 \rangle_{\mathbb{F}_q}^\perp$ and $H_2 = \langle w_2 \rangle_{\mathbb{F}_q}^\perp$, for all $1 \leq i \leq j$, we have $X_i \in \mathcal{X}(H_1)$, while for all $j+1 \leq i \leq n$, we have $X_i \in \mathcal{X}(H_2)$. Therefore

$$\mathcal{X} = \mathcal{X}(H_1) + \mathcal{X}(H_2),$$

meaning that $\mathcal{X}$ is column-hyperplane-decomposable, as claimed. $\square$

Lemma 2.1. *For every subspace $S \subseteq \mathbb{F}_q^k$ of codimension s ,*

$$\dim \mathcal{X}_c(S) \geq \dim \mathcal{X} - sm.$$

Proof. For every subspace $S \subseteq \mathbb{F}_q^k$ of codimension s we have

$$\mathcal{X}_c(S) = \mathcal{X} \cap (S \otimes \mathbb{F}_q^m)$$

so that

$$\begin{aligned} \dim \mathcal{X}_c(S) &= \dim(\mathcal{X} \cap (S \otimes \mathbb{F}_q^m)) \\ &= \dim \mathcal{X} + \dim(S \otimes \mathbb{F}_q^m) - \dim(\mathcal{X} + (S \otimes \mathbb{F}_q^m)) \\ &\geq \dim \mathcal{X} + (k - s)m - km \\ &= \dim \mathcal{X} - sm. \end{aligned}$$

□

Theorem 2.3. *If $\mathcal{X}$ is column-hyperplane-indecomposable, then*

$$\dim \mathcal{X}_c(H) \leq n - \frac{k}{m}$$

for every hyperplane $H \subseteq \mathbb{F}_q^k$. Hence, if $\mathcal{C}$ is a $[m \times n, k, d]_q$ row-intersecting code with $\dim \mathcal{X} = n$, then $m \cdot d \geq k$.

Proof. Let $\dim \mathcal{X}_c(H) = r$ and $X_1, \dots, X_r$ a basis of $\mathcal{X}_c(H)$. Let $X_{r+1}, \dots, X_n$ be $n - r$ linearly independent matrices in $\mathbb{F}_q^{k \times m}$ that, with the matrices $X_1, \dots, X_n$, form a basis for $\mathcal{X}$. If $m \cdot (n - r) \leq k - 1$, there exists a hyperplane H' such that $X_{r+1}, \dots, X_n$ belong to $\mathcal{X}_c(H')$, so that $\mathcal{X}$ is column-hyperplane-decomposable. The second statement is a direct consequence of (1). □

Theorem 2.4.

$$\mathcal{X} \text{ column-hyperplane-indecomposable} \Rightarrow \dim \mathcal{X} \leq 2m - 1.$$

Proof. Let H_1, H_2 two hyperplanes.

$$\begin{aligned} \dim(\mathcal{X}_c(H_1) + \mathcal{X}_c(H_2)) &= \dim \mathcal{X}_c(H_1) + \dim \mathcal{X}_c(H_2) - \dim \mathcal{X}_c(H_1 \cap H_2) \\ &\geq 2(\dim \mathcal{X} - m) - \dim \mathcal{X}_c(H_1 \cap H_2) \end{aligned}$$

So that, if $\dim \mathcal{X}_c(H_1 \cap H_2) = \dim \mathcal{X} - 2m$ then $\mathcal{X}$ is then column-hyperplane-decomposable.

Suppose now $\dim \mathcal{X} = 2m$. Via an isomorphism $\mathbb{F}_q^m \simeq \mathbb{F}_{q^m}$, we may consider $\mathcal{X}$ to be an $\mathbb{F}_q$ -subspace of $\mathbb{F}_q^{k \times m}$, which corresponds to a set of points in $\text{PG}(k - 1, q^m)$ of cardinality at most $\frac{q^{2m} - 1}{q - 1}$. Such a set is too small to meet all $\mathbb{F}_{q^m}$ -subspaces of $\text{PG}(k - 1, q^m)$ of codimension 2 (see [BB66, Theorem 2]). Now, such subspaces correspond, via the isomorphism mentioned before, to the subspaces of $\mathbb{F}_q^{k \times m}$ of type $S \otimes \mathbb{F}_q^m$, with S a subspace of codimension 2. This means that there exists S of codimension 2 such that $\mathcal{X}_c(S) = \{0\}$.

Let H_1, H_2 such that $S = H_1 \cap H_2$. Then $\dim \mathcal{X}_c(H_1 \cap H_2) = 0 = 2m - 2m$, so that $\mathcal{X}$ is column-hyperplane-decomposable.

If $\dim \mathcal{X} > 2m$, it is enough to pick a subspace of $\mathcal{X}$ of dimension $2m$ and repeat the argument, as in [BBMS25]. $\square$

Remark 2.1. Note that the argument used in [BBMS25] to get $2m - 3$ instead of $2m - 1$ does not work. In principle, it may then be possible to construct a column-hyperplane-indecomposable space of larger dimension.

Remark 2.2. Theorem 2.2 and 2.4 summarize as follows, in coding-theoretical terms: let $\mathcal{C} \subseteq \mathbb{F}_q^{m \times n}$ be a rank-metric code of dimension k with generator tensor T and let $\mathcal{X} = \text{ss}_3(T) \subseteq \mathbb{F}_q^{k \times m}$ be the third slice space of T . If $\mathcal{C}$ is row-intersecting, then

$$2\lfloor (k-1)/m \rfloor < \dim \mathcal{X} < 2m.$$

In the theorem, we generalize a result of [BBMS25] to matrix intersecting codes. The result shows that linearity over the extended field is actually superfluous for extending MRD codes and obtaining (row)-intersecting codes. The result is formulated in terms of the third slice space.

Theorem 2.5. Let $2 \leq h \leq \lfloor \frac{m+1}{2} \rfloor$. There exists a column-hyperplane indecomposable $\mathcal{X} \subseteq \mathbb{F}_q^{hm \times m}$ with

$$\dim \mathcal{X} \in \{m, \dots, 2m - 2h + 1\}.$$

Proof. Let $\mathcal{X}' \subseteq \mathbb{F}_q^{hm \times m}$ such that $\dim \mathcal{X}'_c(H) \leq h - 1$ for every hyperplane $H \subseteq \mathbb{F}_q^{hm}$ (which corresponds to a square MRD code – e.g. a Gabidulin code – of parameters $[m \times m, hm, m - h + 1]_q$, see [ABN26]). Let $\mathcal{X}''$ be an $\mathbb{F}_q$ -linear subspace of $\mathbb{F}_q^{hm \times m}$ of dimension $r \leq (hm - 1)m$ and such that $\mathcal{X}' \cap \mathcal{X}'' = \{0\}$. Set $\mathcal{X} = \mathcal{X}' \oplus \mathcal{X}''$. Then $\dim \mathcal{X} = m + r$. We want to prove that $\mathcal{X}$ is column-hyperplane-indecomposable whenever $r \leq m - 2h + 1$. Suppose $r \leq m - 2h + 1$ and assume that there exists a hyperplane H such that $\dim \mathcal{X}_c(H) \geq \frac{1}{2} \cdot \dim \mathcal{X}$. Then

$$\begin{aligned} \dim \mathcal{X}'_c(H) &= \dim(\mathcal{X}' \cap (H \otimes \mathbb{F}_q^m)) = \dim(\mathcal{X} \cap \mathcal{X}' \cap (H \otimes \mathbb{F}_q^m)) \\ &= \dim \mathcal{X}_c(H) + \dim \mathcal{X}' - \dim(\mathcal{X}_c(H) + \mathcal{X}') \\ &\geq \dim \mathcal{X}_c(H) + \dim \mathcal{X}' - \dim \mathcal{X} \\ &\geq \frac{m+r}{2} + m - (m+r) = \frac{m-r}{2} \geq \frac{m - (m - 2h + 1)}{2} \geq h, \end{aligned}$$

which contradicts the assumption above. It follows $\dim \mathcal{X}_c(H) < \frac{1}{2} \cdot \dim \mathcal{X}$ for every hyperplane H , so that $\mathcal{X}$ is column-hyperplane-indecomposable by Proposition 2.1. $\square$

Remark 2.3. *The above result, together with Remark 1.3, guarantees the existence of (row)-intersecting codes of length between $2hm - 1$ and $2m - 2h + 1$ and dimension hm , for $2 \leq h \leq \lfloor \frac{m+1}{2} \rfloor$. Clearly, dimensions that are not multiples of m can also be obtained by considering subcodes.*

To conclude, we want to translate the condition of being column-hyperplane-indecomposable in terms of the dual of the slice space. Let us reformulate in our terms the duality shortening-puncturing which is proved in many previous works on rank-metric codes (see for example [BR17]).

Theorem 2.6. *Let $\mathcal{X} \subseteq \mathbb{F}_q^{k \times m}$ and $L \in \mathbb{F}_q^{l \times k}$ of full rank.*

$$({}_L\mathcal{X})^\perp = {}_A(\mathcal{X}^\perp)_c(\text{rowspan}(L)),$$

where $A \in \mathbb{F}_q^{l \times k}$ is such that $AL^t = I_l$.

Proof. Let us first examine $({}_L\mathcal{X})^\perp$. We have

$$\begin{aligned} ({}_L\mathcal{X})^\perp &= \{N \mid \forall M \in \mathcal{X}, \quad \text{Tr}(N^t LM) = 0\} \\ &= \{N \mid \forall M \in \mathcal{X}, \quad \text{Tr}(N^t (L^t)^t M) = 0\} \\ &= \{N \mid \forall M \in \mathcal{X}, \quad \text{Tr}((L^t N)^t M) = 0\} \end{aligned}$$

Now note that for any $N \in ({}_L\mathcal{X})^\perp \subseteq \mathbb{F}_q^{l \times m}$ we have

$$\text{colspan}(L^t N) \subseteq \text{colspan}(L^t) = \text{rowspan}(L).$$

Note also that $\forall P \in \mathbb{F}_q^{k \times m}$ such that $\text{colspan}(P) \subseteq \text{rowspan}(L)$, there exists $N \in \mathbb{F}_q^{l \times m}$ such that $P = L^t N$. Therefore we can write

$$\mathcal{X}_c^\perp(\text{rowspan}(L)) = \{P \mid \forall M \in \mathcal{X}, \quad \text{Tr}(P^t M) = 0 \text{ and } \text{colspan}(P) \subseteq \text{rowspan}(L)\},$$

and so

$$\begin{aligned} ({}_L\mathcal{X})^\perp &= \{N \mid \forall M \in \mathcal{X}, \quad \text{Tr}((L^t N)^t M) = 0\} \\ &= \{AL^t N \mid \forall M \in \mathcal{X}, \quad \text{Tr}((L^t N)^t M) = 0\} \\ &= \{AP \mid \forall M \in \mathcal{X}, \quad \text{Tr}(P^t M) = 0 \text{ and } \text{colspan}(P) \subseteq \text{rowspan}(L)\} \\ &= {}_A(\mathcal{X}^\perp)_c(\text{rowspan}(L)), \end{aligned}$$

which completes the proof. $\square$

Corollary 2.1. *Let $\mathcal{X} \subseteq \mathbb{F}_q^{k \times m}$. The following conditions are equivalent:*

- a) $\mathcal{X}$ is column-hyperplane-decomposable;
- b) There exists two full-rank matrices $L_1, L_2 \in \mathbb{F}_q^{(k-1) \times k}$ such that

$$\mathcal{X} = \mathcal{X}_c(\text{rowspan}(L_1)) + \mathcal{X}_c(\text{rowspan}(L_2));$$

c) There exists two full-rank matrices $L_1, L_2 \in \mathbb{F}_q^{(k-1) \times k}$ such that

$$\mathcal{X}^\perp = \mathcal{X}_c(\text{rowspan}(L_1))^\perp \cap \mathcal{X}_c(\text{rowspan}(L_2))^\perp;$$

d) There exists two full-rank matrices $L_1, L_2 \in \mathbb{F}_q^{(k-1) \times k}$ such that

$$_{A_1} \mathcal{X} = (_{L_1} \mathcal{X}^\perp)^\perp +_{A_1} \mathcal{X}_c(\text{rowspan}(L_2)),$$

where A_1 is such that $A_1 L_1^t = I_{k-1}$.

Let us conclude with the following remark.

Remark 2.4. *In this long abstract, we have focused on generalizing the results of [BBMS25] to the matrix case, highlighting both analogies and differences. In the extended version, we aim to introduce a richer collection of new constructions and to undertake a deeper and more systematic investigation of the shortening–puncturing duality, as applied to the third slice space of intersecting codes, revealing further structural insights and promising connections. The main difference with respect to the linear case is that the geometric object associated with a matrix code is itself a matrix code. Consequently, one can exploit coding-theoretical properties that have so far remained unexplored, such as puncturing and shortening, applied to the slice space rather than to the original code.*

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